:

LAB 11: Partial Differential Equation

Objective: To numerically obtain the governing PDE of any physical system and solve it numerically.

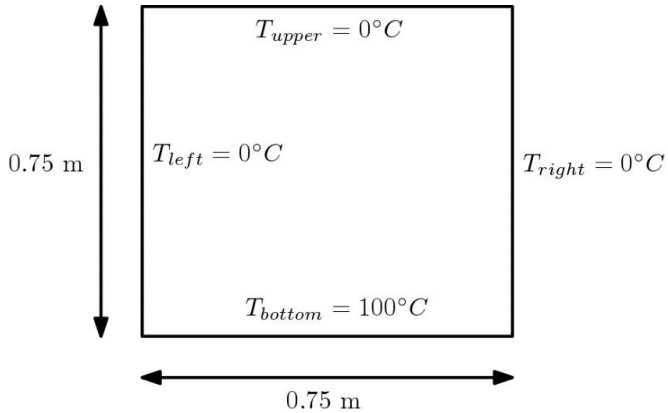
Theory:

Theory - Consider a 2D metal plate with dimensions 0.75x0.75 m2 is kept at 100 degree C at one of it end and rest of the ends are initially at 0 degree C as shown in the figure.

The transient Fourier law of heat conduction for 2D metal plate is given by,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)$$

Where T(x,y,t) is the temperature at any location and time and x, y, t are space and time coordinates. Assume the thermal conductivity of plate is K = 1 W/m degree C.



Take the spatial discretization as delta x = delta y = 0.025 m.

Using explicit finite difference method, the discrete equation can be obtained using Taylor series expansion as below,

$$\frac{T^{n+1}_{i,j} - T^{n}_{i,j}}{\Delta t} = k\left(\frac{T^{n}_{i+1,j} - 2T^{n}_{i,j} + T^{n}_{i-1,j}}{\Delta x^2} + \frac{T^{n}_{i,j+1} - 2T^{n}_{i,j} + T^{n}_{i,j-1}}{\Delta y^2}\right)$$

$$\text{Suppose } \Delta x = \Delta y = \Delta h$$

$$T^{n+1}_{i,j} = T^{n}_{i,j} + \frac{k\Delta t}{\Delta h^2}\left(T^{n}_{i+1,j} - 2T^{n}_{i,j} + T^{n}_{i-1,j} + T^{n}_{i,j+1} - 2T^{n}_{i,j} + T^{n}_{i,j-1}\right)$$

Take the spatial discretization as $\Delta x = \Delta y = \Delta h = 0.025$ m and $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

MATLAB program-

% Explicit method for 2D heat equation in flat plate

```
close all; clear all; clc;
L =0.75;
dx = 0.05;
```

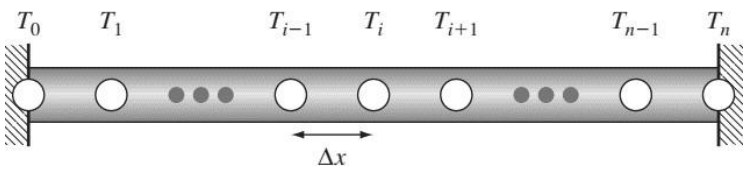
```
dy=0.05;
N = L/dx +1;
x = linspace(0,L,N);
y = linspace(0,L,N);
dt = 0.00015;
epsilon     =     1e-4;
T_new = zeros(N,N);
 %boundary condition
      T_new(1,:) =100;
      T_new(:,1) =0;
      T_new(N,:)=0;
      T_new(:,N)=0;

error =1;iter=0;
while(error > epsilon)
    iter=iter+1;
      T = T_new;% update T every iteration
   for i=2:N-1
      for j =2:N-1
         T_new(i,j) = dt*((T(i+1,j)-2*T(i,j)+T(i-1,j))/dx^2 +(T(i,j+1)-2*T(i,j)+T(i,j-1))/dy^2) +
T(i,j);

      end
   end
   error = max(max(abs(T-T_new)));

figure(2);
hold on;
pause(0.001);
%     contourf(T_new);
    shading flat;colorbar;
    contourf(x,y,T_new,'ShowText','on')
colorbar;
xlabel('x meter');ylabel('y meter')

end
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Consider 1D heat conduction in metal rod as below.



1D heat conduction governing equation is given below,

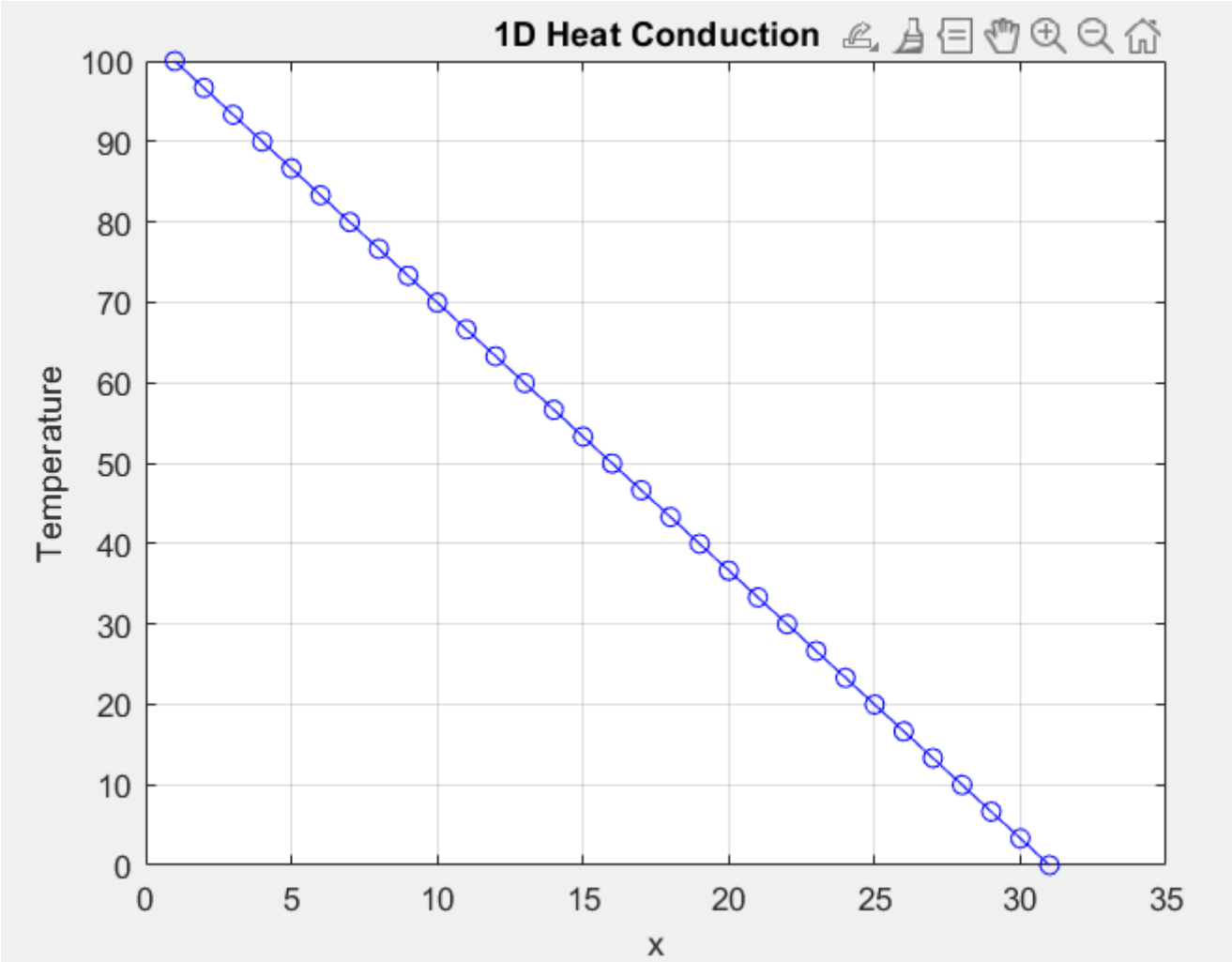
$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2}\right)$$

Write down the discretized equation is explicit finite difference method and obtain the Temperature profile T(x,t). Take length of rod is 0.75 m, Δx= 0.025 m, $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

Code:

```
clc;
clear all;

L = 0.75; dx = 0.025; dt = 0.00015;
K = 1; eps = 1e-4;
N = L/dx + 1;

T = zeros(1,N);
T(1) = 100;   % left end
T(N) = 0;     % right end

error = 1;

while error > eps
   T_old = T;

   for i = 2:N-1
      T(i) = T_old(i) + K*dt/dx^2 * ...
          (T_old(i+1) - 2*T_old(i) + T_old(i-1));
   end

   error = max(abs(T - T_old));
end

plot(T,'b-o'); grid on;
title('1D Heat Conduction');
xlabel('x'); ylabel('Temperature');
```

Result:


1D Heat Conduction
Temperature
x
100
90
80
70
60
50
40
30
20
10
0
0
5
10
15
20
25
30
35